

lecture

November 7, 2025

at dimension n , we can embed any ordinal relationship

$d \leq n$ embedding is always possible

use matrix with epsilons, row vectors is embedding for that entry. computing shows this embedding respects ordinality

$$\frac{n}{2} \leq d \leq n \quad (1)$$

lower bound is proven by saying there are too many comparisons (asymptotically too many), and there aren't enough comparisons in $\mathbb{R}^{n/2}$

the polynomials have degree 2, since the eucl norm squared gives just $x_1^2 + \dots + x_d^2$. there are $n \cdot d$ variables, and there are $\binom{n}{2} \approx n^2$ polynomials.

1 Terminal (VIP) ordinal embeddings

$$T = \{t_1 \dots t_k\}, V \setminus T = \{v_1 \dots v_{n-k}\} \quad (2)$$

you only care about comparing distances of only the terminal nodes, i.e. of the form

$$(t, \cdot) < (t', \cdot) \quad (3)$$

comparisons between vs is irrelevant

this can be done in just k dimensions

assign all t_i to $\vec{e}_i \cdot k^3 n^2$

the vertex $t \in T, v \in V$, rank $r(t, v) \in \{1, 2, \dots, k \cdot n\}$, specifying the order in which distance (t, v) appears w.r.t. other vertices.

to compare (t, v) and t', v' ,

$$|\varphi(t) - \varphi(v)| = \sum_{i \neq t} r(i, v)^2 + ((r(t, v) + M))^2 = \sum_i r^2(i, v) + 2r(t, v)M + M^2 \quad (4)$$

but the first term is $\leq k \cdot (km)^2 \leq M$

2 diameter

diameter of a set of points is max 1 norm between two points.

n is number of points and d is the dimension naive is $O(n^2 d)$, better is $O(nd2^d)$

generate 2^d sign vectors s of form $\prod_d \pm 1$

make f such that $f(p_i) = s_i \cdot p$

f over all p has $n2^d$ entries.

$$|p - q|_1 = |f(p) - f(q)|_\infty \tag{5}$$